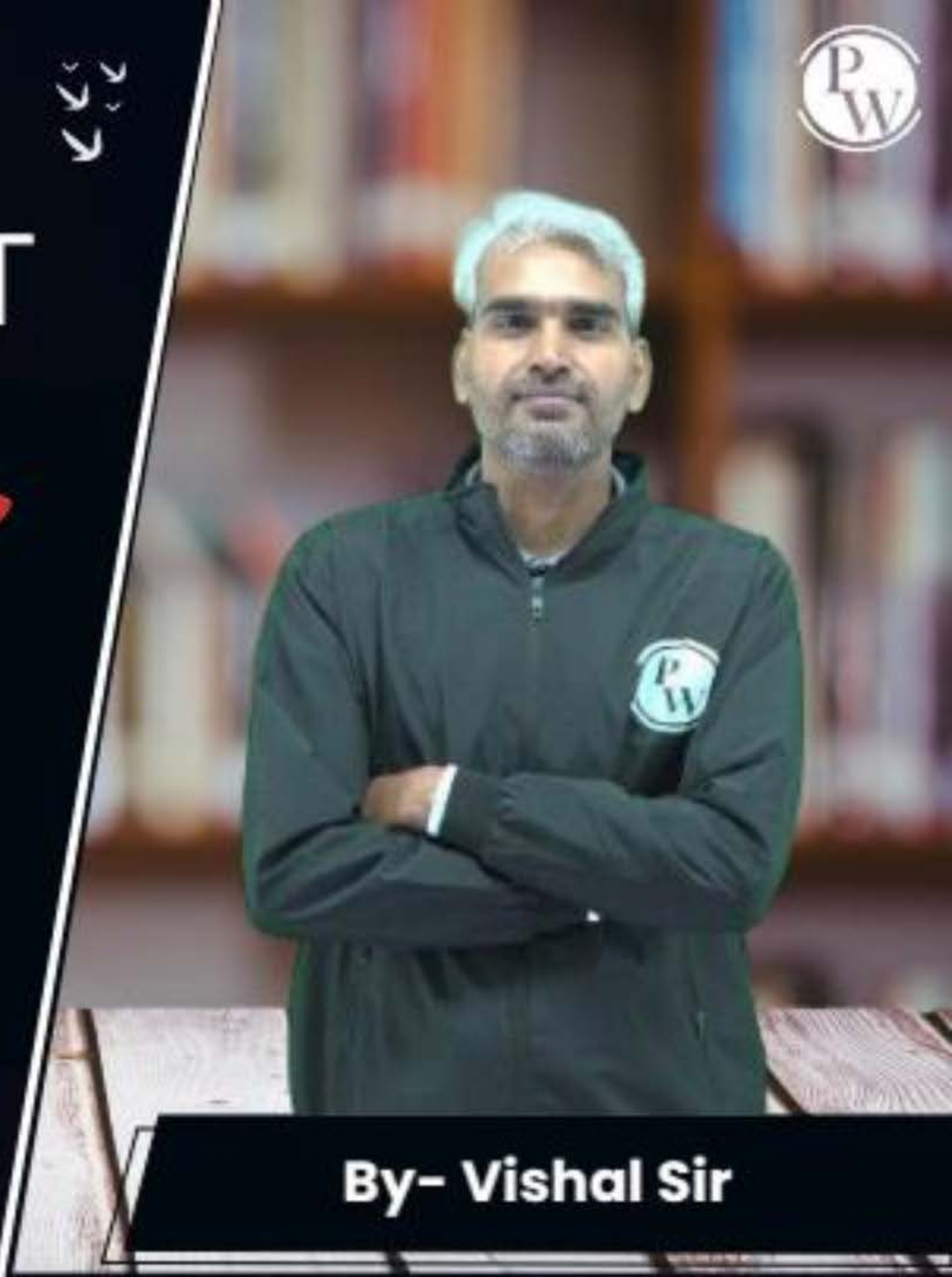
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



Topic — Quick revision

Topic — Havel Hakimi's result

Complement of a graph

Slide

Topics to be Covered



*(2) Topic — Self complementary graph

*(1) Topic — Graph isomorphism

Slide



Topic : Graph Isomorphism

Two graphs are isomorphic if and only if they have exact same properties { Shape & name need not be same }

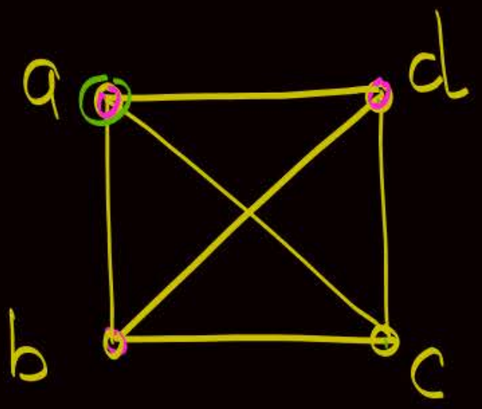
- Two graphs G and G' are said to be isomorphic if there exists a function f: V(G)→V(G') such that

I. f is bijective i.e. $|V(G)| = |V(G')|$

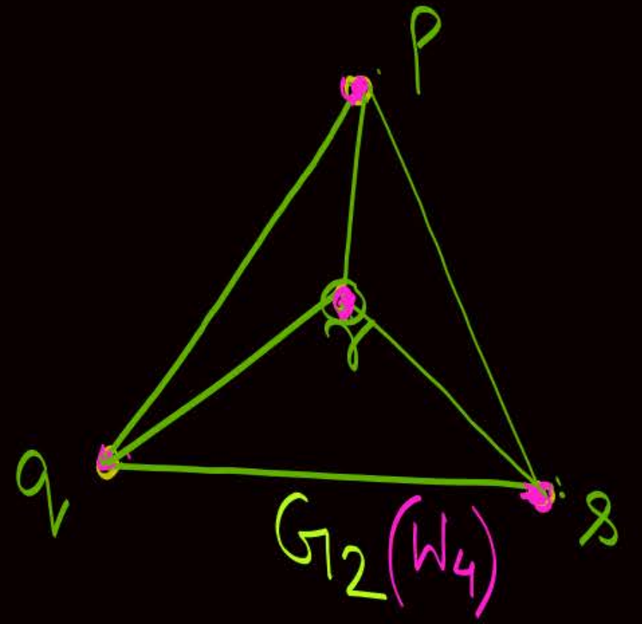
II. f preserves adjacency of vertices

i.e. if two vertices $v_1, v_2 \in V(G)$ are adjacent to each other in graph G. then their images w.r.t. function f i.e. $f(v_1), f(v_2) \in V(G')$ must also be adjacent And Vice-versa.

} ⇒ ∴ $|E(G)| = |E(G')|$



$G_1(K_4)$

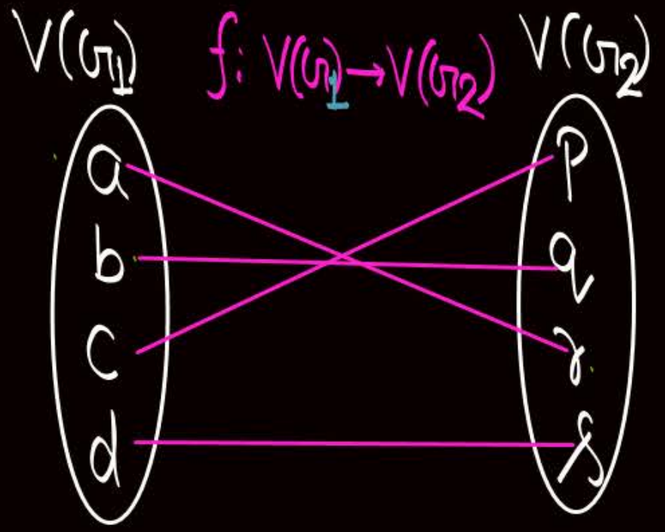


$G_2(W_4)$

Check whether G_1 & G_2 are isomorphic or not ?

$V(G_1) = \{a, b, c, d\}$

$V(G_2) = \{p, q, r, s\}$



$f(a) = r$
$f(b) = q$
$f(c) = p$
$f(d) = s$

① Function $f: V(G_1) \to V(G_2)$ is bijective

& ② 'f' preserves the adjacency of vertices

Hence, G_1 & G_2 are isomorphic

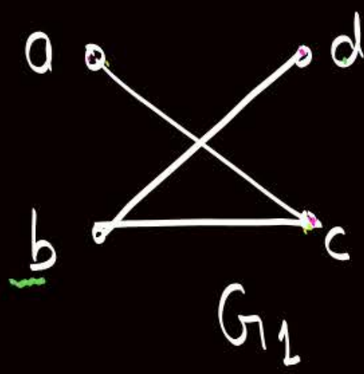
4

G is isomorphic to G' is denoted by

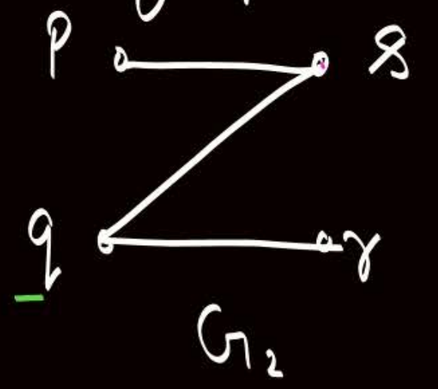
$$G \cong G'$$

"isomorphic"

Q: Consider two graphs G_1 and G_2.
Check whether the graphs are isomorphic or not



G_1



G_2

$|V(G_1)| = |V(G_2)|$

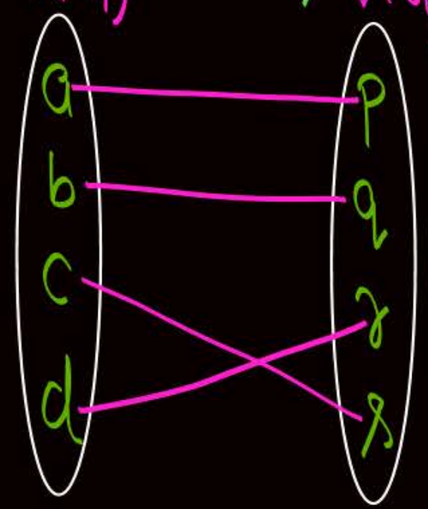
&

$|E(G_1)| = |E(G_2)|$

$V(G_1) = \{a, b, c, d\}$

$V(G_2) = \{p, q, r, s\}$

$f: V(G_1) \longrightarrow V(G_2)$



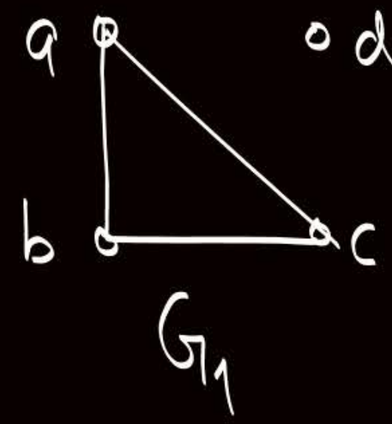
'f' is a bijective function

&

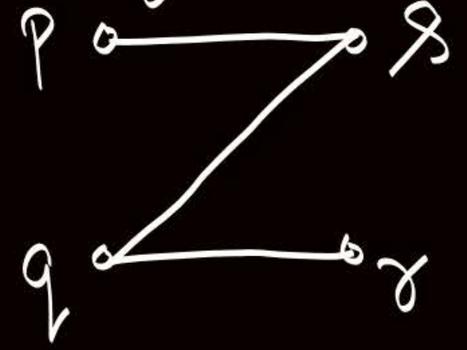
'f' preserves adjacency of vertices

Hence G_1 & G_2 are isomorphic

Q: Consider two graphs G_1 and G_2.
Check whether the graphs are isomorphic or not



G_1



G_2

$|V(G_1)| = |V(G_2)|$

$|E(G_1)| = |E(G_2)|$

Both conditions holds true, but $G_1 \not\cong G_2$ are not isomorphic



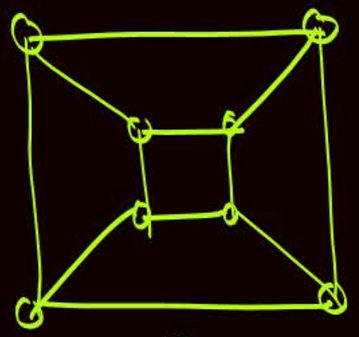
Topic : Graph Isomorphism

If $G \cong G'$, then following conditions must hold true.

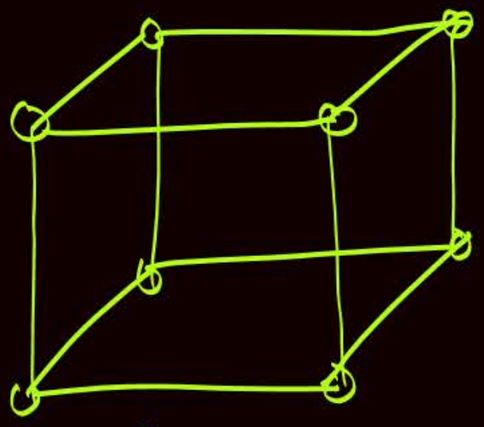
If any of these four conditions is not satisfied, then graphs can not be isomorphic to each other

1. $|V(G)| = |V(G')|$
2. $|E(G)| = |E(G')|$
3. Degree sequence of G = Degree sequence of G'
4. If some 'k' vertices $v_1, v_2, \ldots v_k$ in graph G form a cycle of length 'k', then their corresponding images in G' i.e. $f(v_1), f(v_2), f(v_3), \ldots f(v_k)$ must also form a cycle of length 'k'.

Q: Consider two graphs G_1 & G_2



G_1

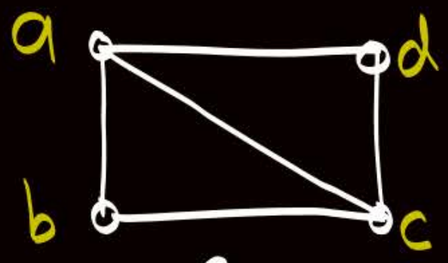


G_2

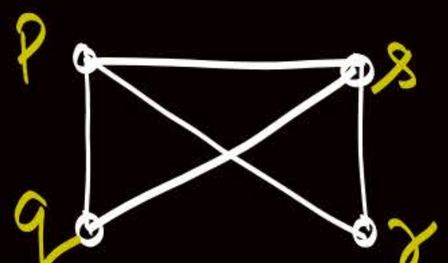
Check whether G_1 & G_2 are isomorphic or not ?

We can observe $G_1 \cong G_2$

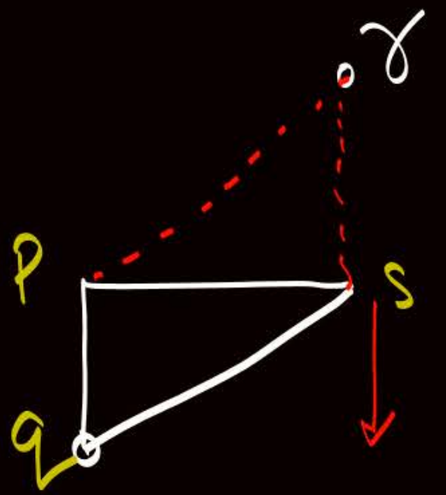
Q: Consider two graphs G_1 & G_2

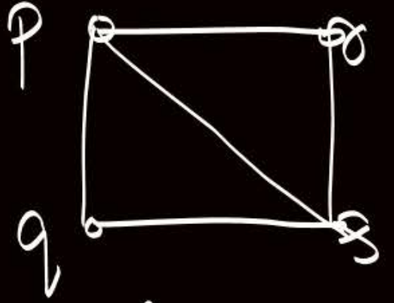


G_1



G_2





same as G_1

Check whether G_1 & G_2 are isomorphic or not?

We can observe

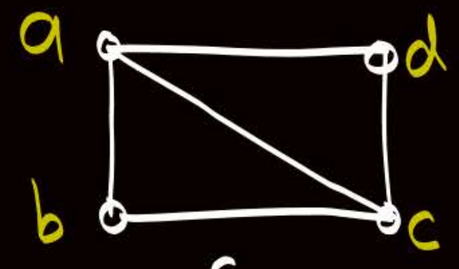
$$G_1 \cong G_2$$



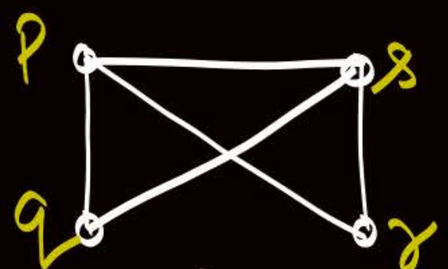
Topic : Graph Isomorphism

Note:- Two simple graphs G_1 and G_2 are isomorphic to each other if and only if their complements are isomorphic to each other

Q:- Consider two graphs G_1 & G_2

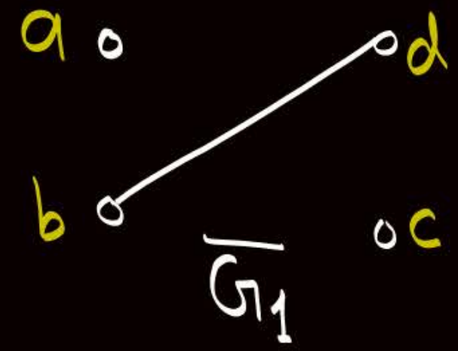


G_1

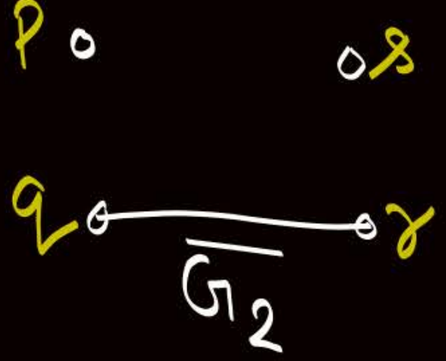


G_2

Check whether G_1 & G_2 are isomorphic or not?



$\overline{G_1}$



$\overline{G_2}$

We can observe that $\overline{G_1} \cong \overline{G_2}$

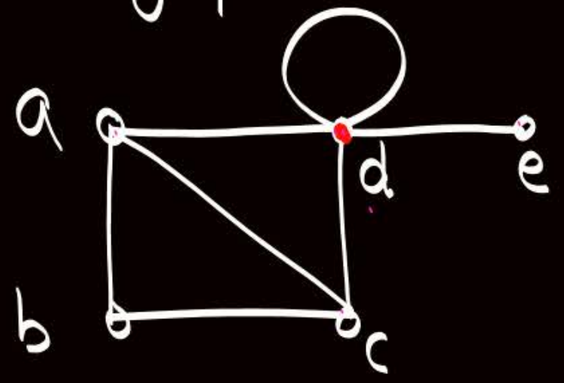
Hence, $G_1 \cong G_2$



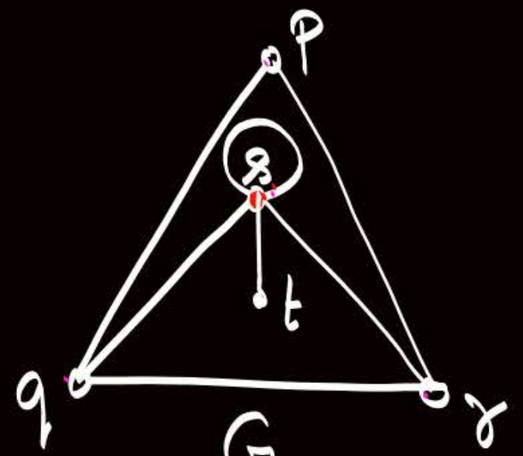
Topic : Graph Isomorphism

* Two graphs G_1 and G_2 are isomorphic if and only if their corresponding subgraphs {i.e the graph obtained by deleting some vertices from graph G_1 and the graph obtained by deleting the the images of those vertices from G_2} are isomorphic to each other

Q: Consider two graphs G_1 and G_2, and check whether the graphs are isomorphic or not?



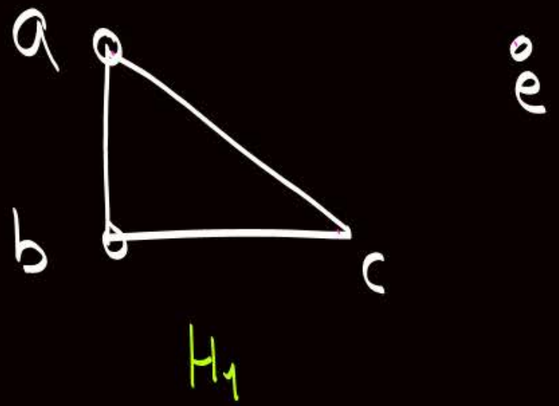
G_1



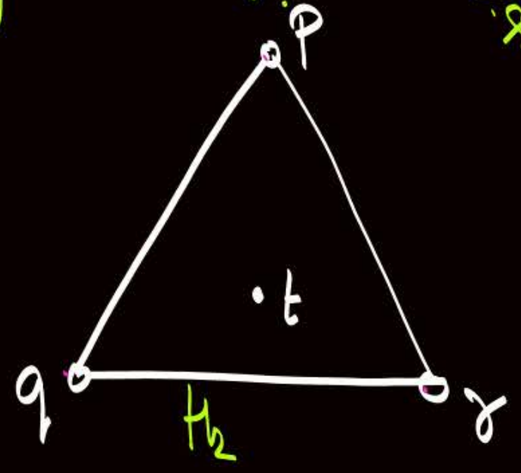
G_2

Properties of d in G_1 are same as properties of s in G_2, ∴ image of d is s

Let H_1 is a sub-graph obtained on deleting 'd' from G_1 and H_2 is a sub-graph obtained on deleting 's'(image of 'd') from G_2



H_1



H_2

} We can observe $H_1 \cong H_2$

$\therefore G_1 \cong G_2$

Q: How many simple graphs are possible with '3' labeled vertices.

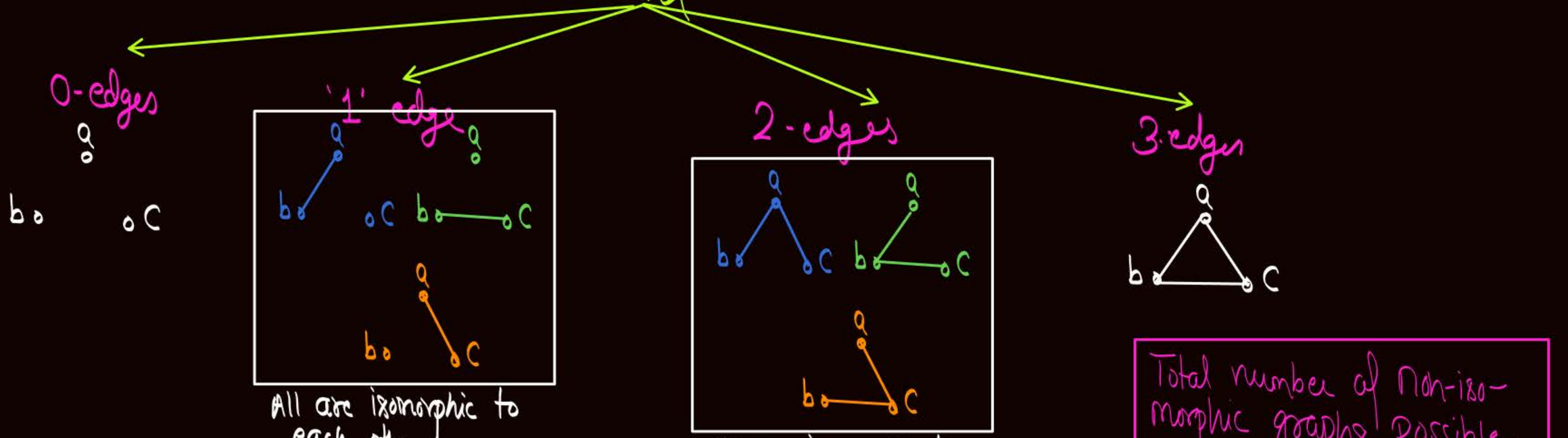
Soln. $(2)^{3C_2} = 2^3 = 8$

Q: How many simple non-isomorphic graphs are possible with '3' labeled vertices

Q: How many simple non-isomorphic graphs are possible with '3' labeled vertices

With '3' vertices, maximum number of Edges possible = $^3C_2 = 3$

Non-isomorphic simple graphs possible with '3' vertices and



All are isomorphic to each other

∴ Counted as a single non-isomorphic graph

All are isomorphic to each other

∴ Counted as a single non-isomorphic graph

Total number of non-isomorphic graphs possible with '3' labeled vertices

$= 1 + 1 + 1 + 1 = 4$

Q:: How many Simple graphs are possible with '4' labeled vertices.

Soln $= (2)^{^4C_2} = 2^6 = 64$

HW Q:- How many Simple "non-isomorphic" graphs are Possible with '4' labeled Vertices.

Q:- How many Simple "non-isomorphic" graphs are Possible with 'n' labeled vertices and two edges.

if 'n=3' : then only one non-isomorphic simple graphs are Possible with '2' edges.

but if $n \geq 4$, then Ans = 2

- '1' when both edges are adjacent to Each other
- '1' when edges are not adjacent to each other

Q: How many simple non-isomorphic graphs are possible with upto '3' vertices.

$n = 1$



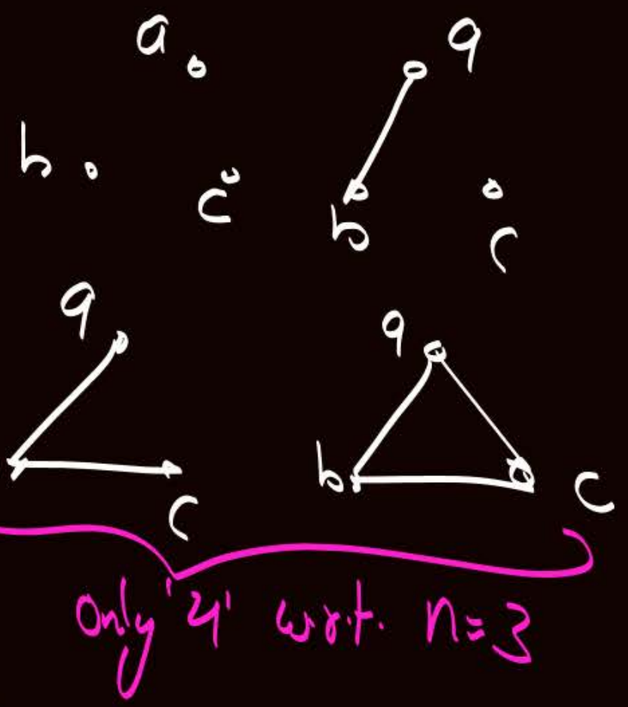
Only '1' w.r.t. $n = 1$

$n = 2$



Only '2' w.r.t. $n = 2$

$n = 3$



Only '4' w.r.t. $n = 3$

Total no. of non-isomorphic graphs possible with upto '3' vertices $= 1 + 2 + 4 = 7$

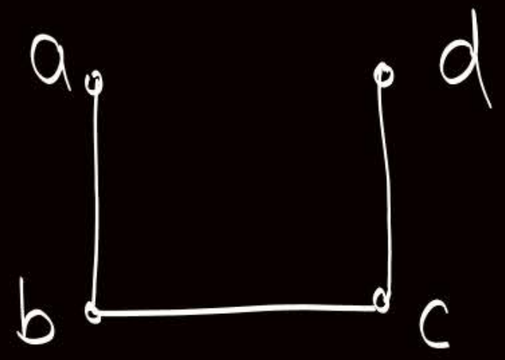


Topic : Self-Complementary Graph

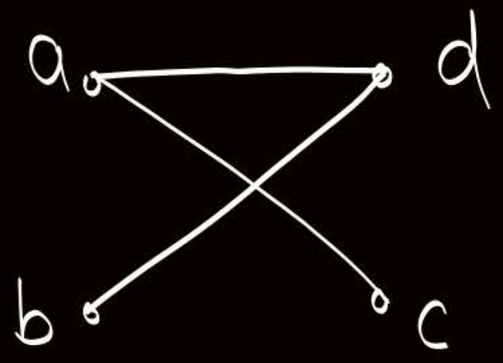
A simple graph G is said to be self-complementary if and only if complement of G is isomorphic to graph G itself.

Graph G is self Complementary if and only if $G \cong \overline{G}$

eg: Let



G



$\overline{G}$

We can observe $G \cong \overline{G}$

i.e graph G is a self-complementary graph

Note:- In a self-complementary graph 'G' with 'n' vertices.

i.e. $G \cong \overline{G}$

① Because $G \cong \overline{G}$

$$\therefore |E(G)| = |E(\overline{G})| \quad \text{——— ①}$$

② G & $\overline{G}$ are Complement of each other

$$\therefore |E(G)| + |E(\overline{G})| = \frac{n(n-1)}{2} \quad \text{——— ②}$$

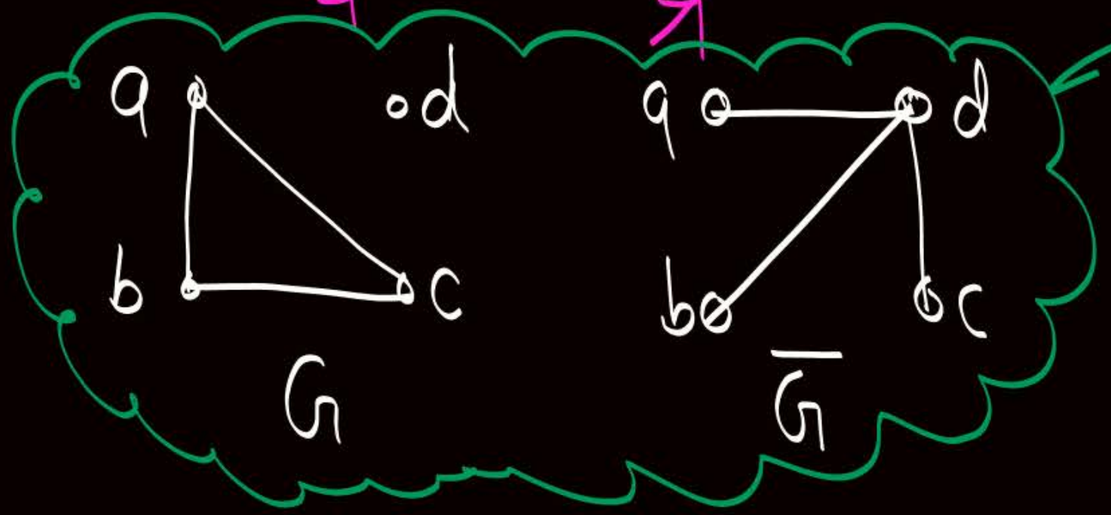
By eq' ① & eq' ②

$$2\,|E(G)| = \frac{n(n-1)}{2} \Rightarrow \boxed{|E(G)| = \frac{n(n-1)}{4}}$$

Note : If G is a self Complementary graph with 'n' vertices then $|E(G)| = \frac{n(n-1)}{4}$, but Converse of the statement need not be true

eg let $n = 4$

$\therefore \frac{n(n-1)}{4} = \frac{4 \cdot (4-1)}{4} = 3$



$|E| = \frac{n(n-1)}{4}$, but graph is not a self-complementary graph

Note: In a self Complementary graph 'G'

$|E(G)| = \frac{n(n-1)}{4}$, and number of edges can never be in fraction.

∴ If graph G is a self Complementary graph then $\frac{n(n-1)}{4}$ must be an integer

∴ either 'n' must be multiple of '4'

or

'n-1' must be multiple of 4

i.e. $n = 4k$ or $n = 4k+1$

where k is any positive integer

Note:

In a self-complementary graph G,

$$|V(G)| \bmod 4 = 0 \text{ or } 1$$

H.W. Q:- Let C_n is a cycle graph with 'n' vertices.
If C_n is a self-Complementary graph {i.e. $C_n \cong \overline{C_n}$}
then Find the value of 'n'.



2 mins Summary

2 Topic Self complementary graph

1 Topic Graph isomorphism

Slide

THANK - YOU